REAL-SILICON-CALIBRATED SIMULATOR · PHASE 0 CHARACTERIZATION

CIM as a Peer Compute Unit for LLM Inference on a Heterogeneous Mobile SoC

From op inventory to silicon: workload statistics and board characterization across Phases 0.1–0.3

Scope dense Llama-3 / Qwen-2.5, 1B-8B, INT8, batch = 1, prefill + decode

Silicon Axelera Metis Alpha (CIM) + production Metis Card **Models** Llama-3.2-1B/3B · Llama-3.1-8B · Qwen2.5-7B

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ABSTRACT

We characterize LLM inference for a *simulated* CIM-enabled heterogeneous mobile SoC, calibrated against two real Axelera Metis boards. Phase 0.1 extracts a device-independent **op inventory** (580 distinct op×shape signatures across 9 categories) from four HuggingFace models under four workloads. Phase 0.2 turns this into a per-(model×workload) **execution profile**: weight-stationary matmul dominates compute (≥0.99 of FLOPs in short/medium workloads; long-context prefill the exception, attention rising to ~0.3), decode is uniformly memory-bound (operational intensity = 2.0 FLOP/byte), and only long-context prefill is attention-memory-bound. Phase 0.3 measures the real silicon: the Compute-in-Memory unit (Metis Alpha, via a weight-stationary 1×1-conv proxy) is fast on matmul but pays a **~911 μs per-call DMA floor** and, for dynamic attention it cannot keep stationary, a **~31–46 ms/token reload penalty — 97–376× slower than GPU-native** — the empirical basis for a CIM-centric design that offloads attention. A micro→end-to-end bridge fit on 1B/3B predicts the held-out 8B decode throughput to **10 %**, with an effective bandwidth (16–21 GB/s) that tracks the production card's documented ~24 GB/s decode wall.

1 Motivation & position

Commercial discrete CIM accelerators (Axelera Metis, Hailo, Mythic, Untether) ship as PCIe/M.2 cards; mobile SoCs integrate NPU + GPU + CPU over shared LPDDR. Their combination — **CIM as a unified-memory-attached peer compute unit on a heterogeneous mobile SoC** — is where the industry is heading, yet is almost absent from the 2024–2026 literature for LLM workloads. Such a chip does not exist, and real Metis cards cannot be the research object directly (the Alpha cannot run LLM compute; the production card runs LLMs only through a closed, precompiled toolchain). We therefore study a *simulated* CIM-mobile SoC and use the real Metis silicon as calibration ground truth.

CIM-centric design discipline. The system is designed *around CIM's limits first* — it excels at weight-stationary GEMV, is poor at dynamic attention, needs channel-aligned tiles, and pays a host–device round trip. GPU/NPU/CPU are the support layer for what CIM cannot or should not do. Crucially, the op→unit assignment is **decided by measurement, not assumed**. This report is that measurement.

2 Method & setup

Three machines, one private repository as the single source of truth. All measurements store raw per-iteration data; every figure in this report is a build artifact regenerable from committed JSON.

Table 1. The three-machine measurement topology.

Machine	Role	Key silicon	SDK
Mac (M3 Pro)	control / dev / git hub	–	–
aetina (RK3588)	CIM + GPU + NPU + CPU micro-bench	Metis Alpha 1 GiB · Mali-G610 · A76	v1.3.1
metiscard (i9-12900K)	end-to-end LLM (L4 anchor)	Metis Card 16 GiB · RTX 3090	v1.6

Table 2. Model configurations (GQA in all four; kv-width = kv_heads × head_dim).

Model	layers	hidden	heads / kv	head_dim	ffn	vocab	tied
Llama-3.2-1B	16	2048	32 / 8	64	8192	128256	yes
Llama-3.2-3B	28	3072	24 / 8	128	8192	128256	yes
Llama-3.1-8B	32	4096	32 / 8	128	14336	128256	no
Qwen2.5-7B	28	3584	28 / 4	128	18944	152064	no

Table 3. Workloads (Layer-A), spanning the prefill-heavy ↔ decode-heavy spectrum. Mean prefill/decode token counts from Phase 0.1 tokenization (Llama family; Qwen ≈ same).

Task	Dataset	prefill	decode	regime
Long-context	LongBench-TriviaQA	11753	4	prefill-heavy
Code	HumanEval	132	54	moderate
QA / reasoning	GSM8K	59	102	balanced
Chat	ShareGPT	175	344	decode-heavy

> Validation layers cross-checked: L1 CIM per-op, L2 DRAM/PCIe, L3 NPU/GPU per-op, L4 end-to-end LLM (production card), L5 sensitivity, L6 end-to-end CNN (reused). This report covers L1–L4; the op→unit mapping is characterization-driven.

3 Phase 0.1 — op inventory

Using a PyTorch meta/FakeTensor + dispatch tracer with forced eager attention, we extract the exact op×shape stream each model executes — device-independent, no GPU, no board. All four models decompose into the same ~38 ATen primitives; the only op-set difference is Qwen's biased QKV (`addmm` vs Llama's `mm`). A completeness cross-check and a real-weights 1B run both pass. The deduplicated **sweep matrix** is 580 distinct `(op, in_shapes, out_shape)` signatures, categorized by semantic origin into 9 classes: matmul 105, attention 95, rope 190, norm 90, softmax 21, ffn 30, residual 20, embedding 20, kv_cache 9. This is the op set Phases 0.2–0.3 weight and measure.

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4 Phase 0.2 — workload-op statistics

Phase 0.2 attaches *weights* to the sweep matrix: for every signature, how many times it executes in each (model × workload), plus FLOPs / bytes / operational intensity from shape. Counts come from the Phase-0.1 inventory (the counting oracle — never hand-rolled); off-grid workload-length shapes are generated by a data-driven length-template validated to reproduce held-out inventory points exactly for all four models. Intensity is the predicted side of the roofline; the measured knee is Phase 0.3.

≥0.99	**2.0**	**65 %**
matmul share of prefill+decode FLOPs (short/medium tasks; LongBench 0.61–0.74)	decode matmul operational intensity (FLOP/byte) – memory-bound everywhere	LongBench prefill *memory* that is attention ($O(S^2)$ score matrix), not weights

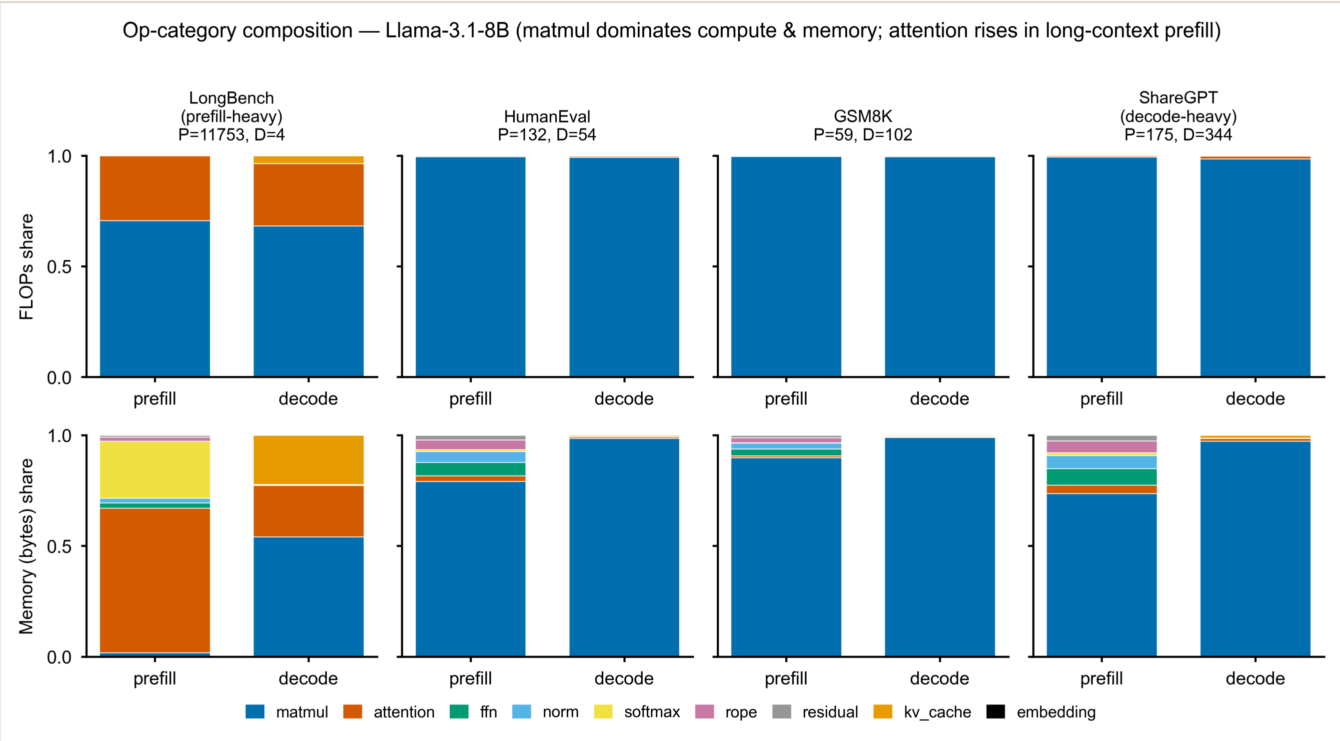

Figure 1. Op-category composition for Llama-3.1-8B across the four workloads, by FLOPs (top) and memory bytes (bottom), prefill vs decode. Weight-stationary **matmul** dominates compute and (in decode) memory everywhere; the exception is long-context prefill (LongBench), where the $O(S^2)$ **attention** score matrix and softmax dominate *memory*. This is the structural fit for a CIM-centric design, and the one regime where a weights-only memory model is wrong.

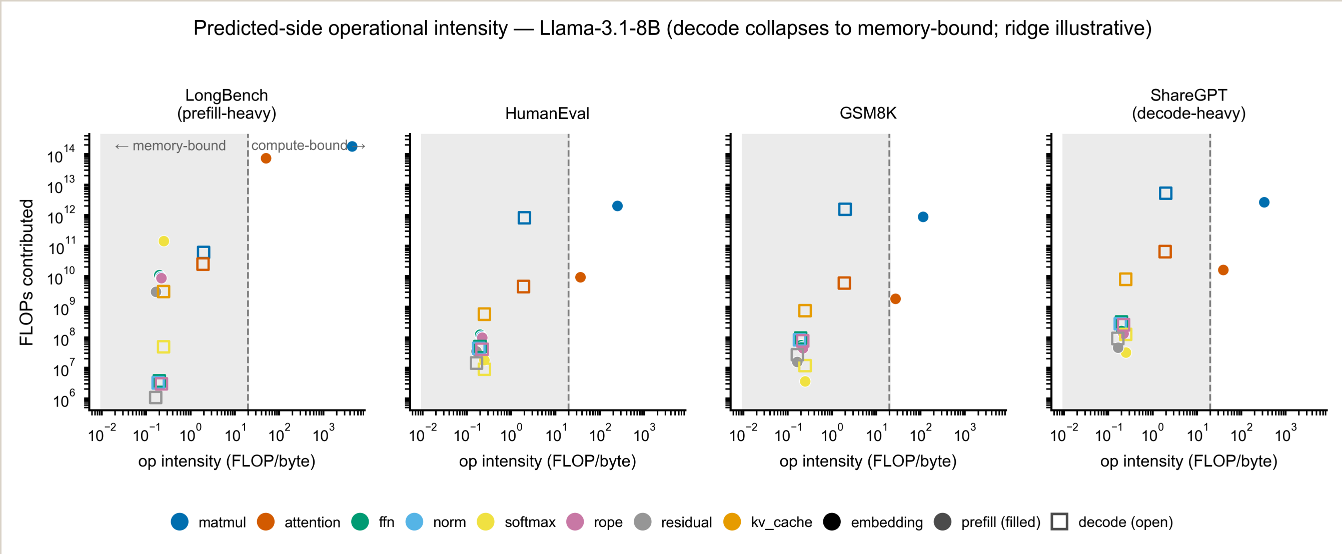

Figure 2. Predicted-side operational intensity (FLOP/byte) vs FLOPs contributed, prefill (filled) vs decode (open). Decode collapses to the memory-bound region (intensity ≈ 2, M = 1 GEMV) – the regime where CIM weight-residency helps most – while prefill matmul/attention is compute-bound and scales with context length (matmul intensity 115 → 5554 from GSM8K to LongBench).

5 Phase 0.3 — real-board characterization

We now measure the silicon. CIM is the Metis Alpha via a 1×1-conv proxy — faithful for weight-stationary matmul/GEMV (the conv filter is a compile-time-constant weight), a compute lower bound for attention (both operands are activations). A hard device-memory envelope (`K·N ≤ ~6M` params allocatable) means large projections run as 2048×2048 crossbar tiles; we measure one canonical tile and extrapolate `n_tiles × tile_latency` — itself the real CIM tiling cost.

911 μs	97-376×	10 %
per-call host↔device DMA floor – dominates small decode GEMVs (A1d.5)	CIM composed attention slower than GPU-native → offload (C4)	8B decode tok/s hold-out error from a 1B/3B fit (C5)

5.1 CIM is fast on matmul but gated by the round trip

Decode-GEMV *device* throughput is 110–230 GFLOP/s, but the per-call DMA floor makes the *end-to-end* latency of a tiny decode GEMV ~0.9 ms regardless of its compute (Figure 3). This on-PCIe floor is the structural CIM cost: CIM wins on compute, loses on the round trip — and tiling a large op multiplies the floor by `n_tiles`.

Table 4. CIM micro-characterization (Metis Alpha, 1×1-conv proxy, INT8, decode M = 1).

Axis	Result
Device envelope	K·N ≤ ~6M params allocatable (6.3M ok, 8.4M+ fail); larger ops tiled 2048×2048
A1d.2 channel staircase	dev latency 9.8 → 24.7 → 41.2 → 82.4 μs for N = 64, 1024, 2048, 3072
A1d.3 (M,K,N) aspect	equal-MAC wide / tall 227, square 204 GFLOP/s (~10% aspect penalty)
A1d.4 l2 vs ddr	1.00–1.01× – *null axis* (Alpha has no on-card DRAM; not extrapolable)
A1d.6 GQA-narrow waste	kv-proj (N = 512) 112 vs wide gate/up 204 GFLOP/s – ~55% utilization (1B)

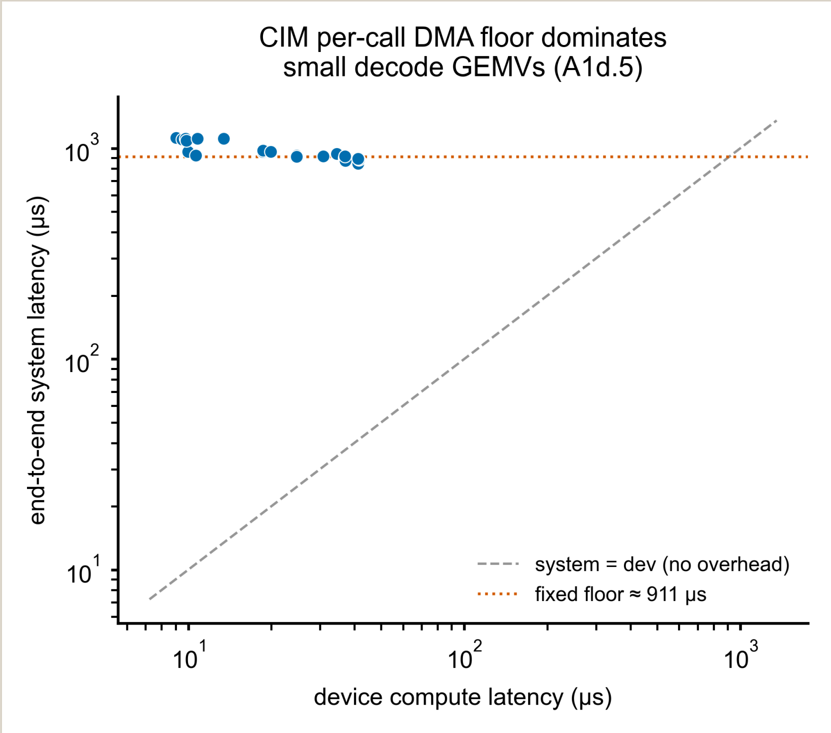

Figure 3. CIM end-to-end system latency vs device compute latency across shapes. The ~911 µs per-call DMA floor (orange) sits far above the no-overhead diagonal – small decode GEMVs are floor-bound, not compute-bound. This is the host-device cost the simulator's memory layer (M2) must model.

5.2 CIM cannot keep attention stationary – the offload argument

The conv-proxy attention compute floor is tiny (~20 μs single-head), but it ignores the per-decode-step reload of the growing KV-cache into the crossbar. Composing the measured reload (`T = floor + layers · DMA_floor`) gives ~31–46 ms/token, **99 % of it the per-layer reload** — an Alpha-topology *upper bound*, but the order of magnitude is the point: CIM is 97–376× slower than GPU-native attention (Figure 4). Dynamic activation×activation attention violates CIM's weight-stationary premise; it must be offloaded. This is the empirical heart of the CIM-centric thesis.

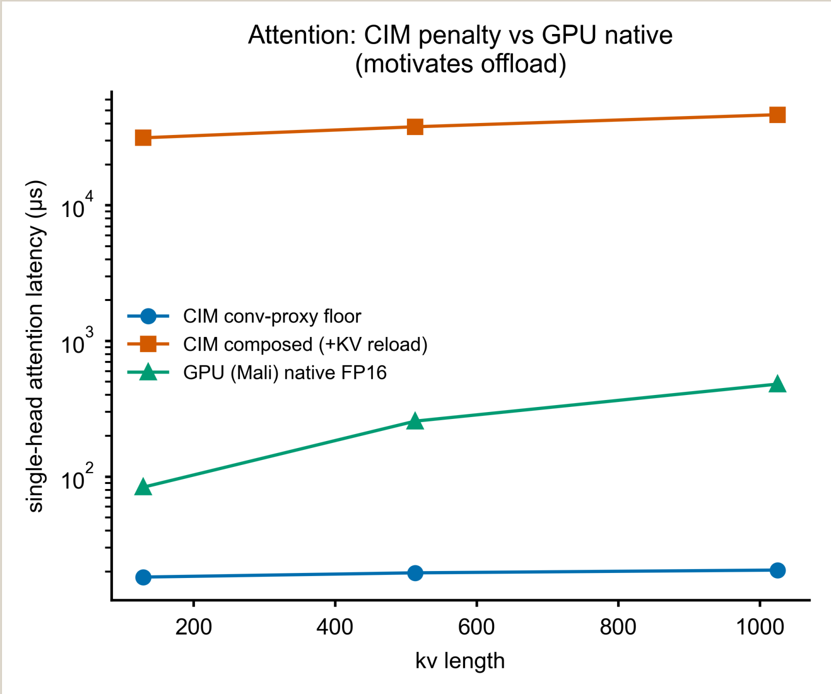

Figure 4. Single-head attention latency vs kv length: CIM conv-proxy **floor** (compute only, misleadingly fast), CIM **composed** (+KV reload, the real ~31–46 ms penalty), and **GPU-native FP16** (Mali, 80–500 µs). The two-order-of-magnitude gap motivates assigning attention to GPU/NPU.

5.3 Micro → end-to-end: predicting the production card

The production Metis Card runs the vendor INT8 LLMs end-to-end (the L4 anchor). Decode is a near-pure weight-streaming memory wall, so we fit an effective decode bandwidth on 1B + 3B and predict the *held-out* 8B throughput (an ADR-0006 size-invariance test). The prediction lands at 2.44 vs measured 2.70 tok/s — 10 % error — and the implied bandwidth (16.2 / 20.5 / 20.3 GB/s) shows a genuine size trend bracketing the ~24 GB/s wall (Figure 5). Op-level structure plus Phase-0.2 counts compose into the real LLM's throughput: the L1→L4 link holds.

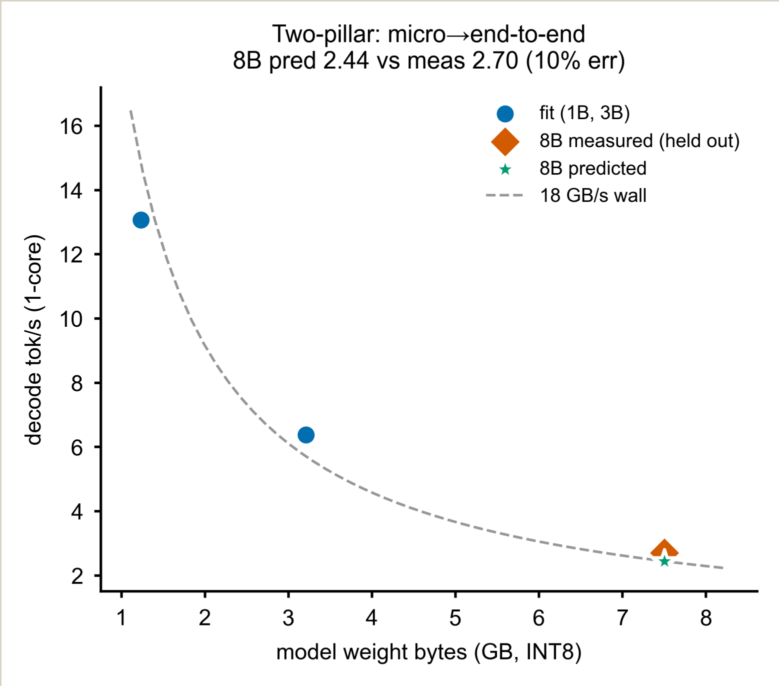

Figure 5. Two-pillar hold-out. Effective decode bandwidth fit on **1B and 3B** predicts the held-out **8B** decode tok/s within 10 % of the **measurement**. The residual is real size-dependence (smaller models stream weights less efficiently), not a fit artifact.

Table 5. Production Metis Card — vendor INT8 decode (axllm, ctx 1024, median of 3 prompts). The 4c/1c speedup shrinks with model size (size-invariance drift), which is why C5 is a hold-out.

Model	1-core tok/s	4-core tok/s	4c/1c	TTFT (1c)	weight (GB)
Llama-3.2-1B	13.07	14.77	1.13×	0.55 s	1.24
Llama-3.2-3B	6.38	6.99	1.10×	1.13 s	3.21
Llama-3.1-8B	2.70	2.92	1.08×	3.79 s	8.03

6 Synthesis — what the silicon says about the design

The measurements substantiate the CIM-centric position with real numbers, not assumptions:

- **Put weight-stationary GEMM/GEMV on CIM.** Matmul is ≥0.99 of prefill and decode FLOPs for short/medium workloads (long-context prefill the exception, attention 0.26–0.39); decode is uniformly memory-bound (intensity 2.0), exactly where CIM weight-residency pays off — provided the per-call DMA floor is amortized (large ops, batched calls).
- **Offload attention.** CIM's ~31–46 ms/token attention penalty (97–376× GPU) is the quantified cost of dynamic activation×activation on a weight-stationary crossbar. The heterogeneous split is not a design choice but a measured necessity.
- **Mind the tile and the round trip.** Channel-64 alignment, GQA-narrow underfill (55 % utilization), and the 911 μs floor are the CIM-specific knobs the simulator's M1/M2 must model.
- **The bridge is real.** A bytes-driven, op-count-weighted model fit on small models predicts the large model's measured throughput to 10 % — evidence the per-op calibration composes into faithful end-to-end behavior.

7 Gaps & next steps

- **RKNPU2 (NPU) not collected** this pass; the attention-offload argument rests on the GPU comparison alone. The C4 composed attention is an Alpha-topology upper-bound estimate; the Mali GEMM kernel is unoptimized (a throughput lower bound).
- **l2/ddr is a null axis on Alpha** (no on-card DRAM) — residency sensitivity must come from the production card, not extrapolated from the Alpha gap.
- **Large-M prefill** (≥2048) exceeds the device allocation envelope; LongBench M ≈ 11.8k stays analytic for Phase 1.
- **Thermal (Phase 0.4):** `core_temp` is unsupported on the production PCIe-Rev1 card; thermal characterization will use the Alpha RK3588 thermal zones.
- **Next:** Phase 1 fits closed-form latency equations per micro-benchmark (replacing lookup tables) and validates each simulator component; Phase 2 integrates them into the end-to-end event-driven simulator.